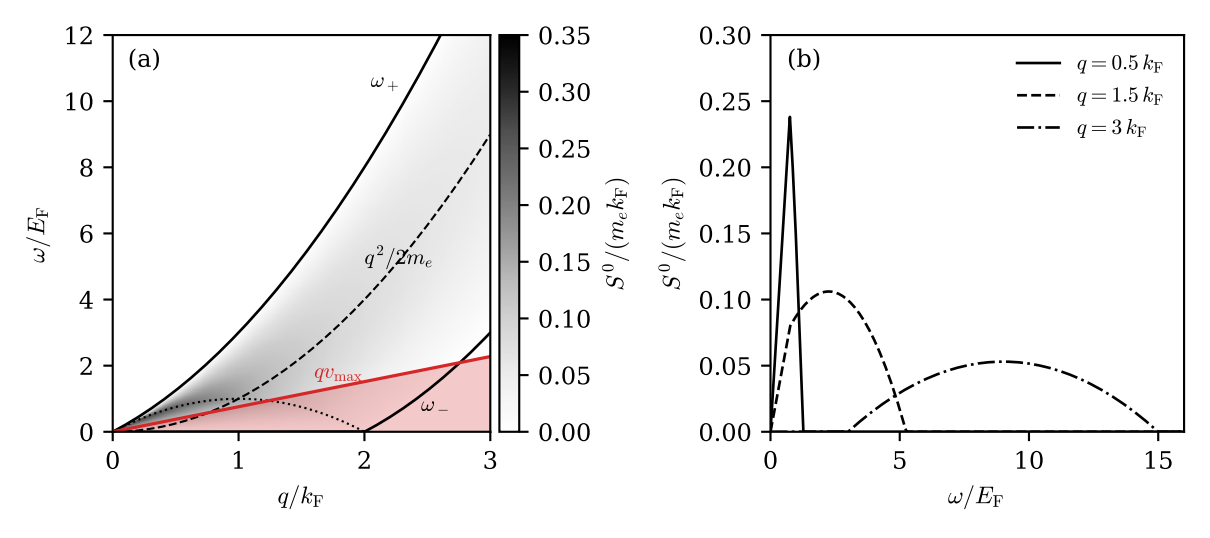
(a)
ω_+
$q^2/2m_e$
$qv_{\max}$
ω_-
$\omega/E_{\rm F}$
$q/k_{\rm F}$
$S^0/(m_e k_{\rm F})$
(b)
$q = 0.5\,k_{\rm F}$
$q = 1.5\,k_{\rm F}$
$q = 3\,k_{\rm F}$
$S^0/(m_e k_{\rm F})$
$\omega/E_{\rm F}$